



Department of Electrical and Computer Engineering
Second Semester, 2025-2026
Second Assignment, Due: June 30, 2026
Arabic Sentiment Analysis

1. Description

This project aims to conduct a comprehensive study of sentiment analysis techniques for Arabic social media data by investigating different text representation methods and classification models. The primary focus is to analyze how various feature representations, including traditional and embedding-based approaches, influence sentiment classification performance, and to compare the effectiveness of non-sequential and sequential deep learning architectures.

Students will explore multiple word representation techniques such as TF-IDF, Word2Vec, FastText, and transformer-based embeddings, and evaluate their impact when combined with different machine learning and deep learning classifiers. The project also emphasizes systematic hyperparameter optimization and rigorous experimental evaluation to identify the most effective combinations of representations and models for Arabic sentiment analysis.

2. Objectives

The primary objectives of this project are:

1. To study and compare different text representation techniques for Arabic sentiment analysis, including:
 - TF-IDF representations
 - Word2Vec embeddings
 - FastText embeddings
 - Contextual embeddings derived from transformer-based language models (optional)
2. To investigate the effectiveness of different classification paradigms by comparing:
 - Traditional machine learning classifiers such as Naïve Bayes and Random Forests.
 - Non-sequential deep learning models such as Convolutional Neural Networks (CNNs).
 - Sequential deep learning models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs).
3. To perform systematic hyperparameter optimization for all models and analyze the effect of model parameters on classification performance.
4. To evaluate and compare all models using standard sentiment analysis metrics including Accuracy, Precision, Recall, and F1-Score.
5. To identify the most effective combinations of text representation methods and classification models for Arabic sentiment classification and provide insights into their strengths and limitations.

3. Dataset Overview.

The project will utilize the attached Arabic sentiment dataset. The data are annotated into three sentiment categories, namely *Positive*, *Negative*, and *Objective/Neutral*. Notably, the dataset presents several challenges, including class imbalance and the presence of linguistic noise such as spelling variations, informal expressions, and dialect-specific vocabulary, all of which must be carefully addressed during the model development and evaluation stages.

4. Tasks and Methodology

1. Data Preprocessing

Perform appropriate Arabic text preprocessing, which may include:

- Text cleaning and normalization.
- Removal of URLs, HTML tags, punctuation, and unnecessary symbols.
- Arabic character normalization.
- Stop-word removal.
- Stemming or lemmatization.
- Handling emojis, hashtags, and negation expressions.

Students should justify the preprocessing pipeline based on the selected representation methods and experimental objectives.

2. Feature Representation and Embedding Analysis

Implement and compare multiple text representation approaches, including:

- TF-IDF
- Word2Vec
- FastText
- Transformer-based embeddings (optional)

Analyze the advantages and limitations of each representation method and their impact on downstream classification performance.

3. Model Development and Training

Train and evaluate the following categories of classifiers:

A. Traditional Machine Learning Models

- Naïve Bayes
- Random Forests

B. Non-Sequential Deep Learning Models

- Convolutional Neural Networks (CNNs)

C. Sequential Deep Learning Models

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)

4. Dataset Splitting and Hyperparameter Optimization

Split the dataset into:

- 60% Training
- 20% Validation
- 20% Testing

Perform systematic hyperparameter tuning for each model.

5. Comparative Evaluation and Analysis

Evaluate all models using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Conduct a comparative analysis that investigates:

- The impact of different text representation methods.
- The performance differences between traditional machine learning and deep learning approaches.
- The performance differences between non-sequential models (CNN) and sequential models (RNN, LSTM).
- The influence of hyperparameter tuning on model performance.
- The strengths and limitations of each representation-model combination.

5. Tools

Python is recommended as the primary programming language for this project. Students may use any suitable libraries and frameworks required to implement and evaluate their models. Commonly used libraries include:

- **Pandas** for data manipulation and analysis.
- **NumPy** for numerical computations.
- **NLTK**, **Camel Tools**, or similar libraries for Arabic text preprocessing.
- **Scikit-learn** for traditional machine learning models, feature extraction, and evaluation.
- **Gensim** for training and utilizing Word2Vec and FastText embeddings.
- **TensorFlow/Keras** or **PyTorch** for implementing deep learning models such as CNNs, RNNs, and LSTMs.
- **Matplotlib** and **Seaborn** for visualization and experimental analysis.

Students may use additional libraries when appropriate, provided they clearly document their usage.

6. Submission

Students are required to submit the following components:

1. Report (PDF Format)

The report should provide a comprehensive description of the entire project and must include the following:

- A clear problem formulation and project overview.
- A description of the dataset and any challenges associated with Arabic sentiment analysis.
- A detailed explanation of the preprocessing pipeline and the rationale behind the selected preprocessing techniques.
- A detailed description of each text representation method investigated, including TF-IDF, Word2Vec, FastText, and any additional embedding methods used.
- A description of all classification models implemented
- A detailed explanation of the architecture and design choices for each deep learning model
- A description of the hyperparameter optimization process
- Experimental results for all representation-model combinations.
- Performance evaluation using appropriate tables, graphs, and visualizations
- A comparative analysis discussing
- A discussion of challenges encountered during the project and the strategies used to address them.

2. Source Code

- Submit all source code electronically. No hard copies are required.
- The code must be well-structured, modular, documented, and executable.
- The submission should include all scripts, notebooks, configuration files, and instructions necessary to reproduce the reported results.
- The code should clearly separate preprocessing, feature representation, model training, hyperparameter tuning, and evaluation components.

Good Luck